PRATIK MANDANI

HYBRID / PRODUCT DEVELOPMENT ENGINEER

Electronics + Embedded Firmware + Web Backend

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Portfolio: pratik-mandani.github.io • GitHub: github.com/pratik-mandani/pratik-mandani

PROFESSIONAL SUMMARY

Hybrid / Product Development Engineer with 7 years of hands-on experience across electronics, vending-machine product development, hardware integration, machine testing, R&D and web application development. Started with hands-on production and electronics work, progressed into product development and R&D, and later expanded into PHP/Laravel full-stack development. Currently strengthening embedded firmware capabilities through personal ESP32 and C/C++ projects, including KOTL — a smart embedded assistant robot.

TECHNICAL SKILLS

PRODUCT DEVELOPMENT & ELECTRONICS

Product Development, Vending Machine Systems, Machine Assembly, Machine Wiring, Hardware Integration, Hardware Testing, Product Testing, PCB Prototyping, PCB Design, PCB Repair & Troubleshooting, Component-Level Troubleshooting, SMD Soldering, Through-Hole Soldering, Electronics, Proteus 8

EMBEDDED & FIRMWARE

C, C++, ESP32, Arduino, GPIO, ADC, I2C, SPI, UART, I2S, Sensor Interfaces, Embedded Firmware Development

WEB & BACKEND

PHP, Laravel, Blade, MySQL, REST API, CRUD, Authentication, Middleware, Sessions, Admin Panels, E-commerce, Business / ERP Applications, HTML, CSS, JavaScript, Bootstrap

TOOLS & DEPLOYMENT

Git, VS Code, FTP, Hostinger, Proteus 8

PROFESSIONAL EXPERIENCE

Altius Infoway — Rajkot, Gujarat

May 2022 – Present (4 Years)

Career Progression: Frontend Intern → Frontend Development → PHP/Laravel Development → Laravel Full-Stack Developer

- Developed and contributed to 100+ websites and web applications across corporate, e-commerce and business domains.
- Built backend applications using PHP and Laravel.
- Worked with MySQL databases and database-driven applications.
- Developed and integrated REST APIs.
- Worked on business management and ERP-style applications.
- Worked with clients on requirements, changes and issue resolution.
- Used Git and FTP in development/deployment workflows.
- Developed responsive frontend interfaces using HTML, CSS, JavaScript and Bootstrap.
- Developed CRUD modules and admin panels.
- Worked with authentication, middleware and sessions.
- Worked on e-commerce functionality.
- Performed bug fixing, maintenance and feature development.
- Managed website deployment, hosting and maintenance.

Symbian Vending Solutions — Rajkot, Gujarat

Nov 2018 – Mar 2022 (3 Years)

Career Progression: Assembly Worker → Production Team Leader → R&D Team Leader

Role: Product Development / R&D / Electronics / Machine Integration • Team: Approx 15 members

- Worked on vending-machine assembly, wiring, testing and operational validation.
- Coordinated procurement, dispatch and production requirements.
- Worked on PCB prototyping and electronics troubleshooting.
- Performed PCB repair and component-level troubleshooting.
- Coordinated with embedded, PLC, Raspberry Pi and application/web development teams during product integration.
- Reported product, production and technical issues to management.
- Coordinated day-to-day production activities with a team of approximately 15 members.
- Provided technical support for machine-related issues.
- Performed SMD and through-hole soldering.
- Worked on vending-machine product integration and R&D.
- Identified machine-level issues during testing and supported debugging and resolution.

PRODUCT DEVELOPMENT PROJECTS — SYMBIAN VENDING

Reusable Vending Machine Control Card

PCB Prototype | Tool: Proteus 8

A small PCB prototype designed to make the vending control system adaptable across machines using common components and configurable connectors.

Responsibilities: PCB prototyping, connector design, hardware integration, testing and troubleshooting.

24V–12V Coin Mechanism Interface & Isolation Module

PCB / Power & Interface Module | Tool: Proteus 8

A custom interface module developed to integrate a 12V coin mechanism into a 24V vending-machine system.

Design: LM2576 power section for 24V → 12V conversion, Relay for 12V switching/control, Optocoupler for signal isolation between the coin mechanism and vending controller.
Tool: Proteus 8

Touch / QR / Automatic Cup Vending Systems

Product Development / Machine Integration

Worked on vending-machine product development involving machine assembly, wiring, hardware integration, operational testing and coordination with embedded, application, PLC, Raspberry Pi and web development teams.

Responsibilities: Machine assembly, wiring, hardware integration, operational testing, developer coordination, R&D support.

PERSONAL EMBEDDED PROJECT

KOTL — The Smart Emotional Assistant Robot

Active Personal Project

A personal embedded hardware and firmware project built to strengthen practical MCU, display, audio and sensor development skills.

Hardware: ESP32, SSD1306 OLED, MAX9814 Microphone, MAX98357A Audio Amplifier, Speaker.
Firmware / Interfaces: C/C++, GPIO, ADC, I2C, I2S.
Current Implementation: OLED RoboEyes-style interface, blinking, idle movement, sound detection, microphone input, audio playback and non-blocking audio playback.

SELECTED WEB PROJECTS — ALTIUS INFOWAY

1. Diamorivo: Laravel + MySQL — E-commerce platform. Frontend UI implemented from Figma design. Frontend + Backend.

2. The Busa: Laravel + MySQL — Fashion e-commerce platform. Frontend + Backend. Product CRUD, Cart, Order Management.

3. Network Kitchenware: Laravel + MySQL — Business management application. Dashboard, User CRUD, Authentication, Database integration.

4. Luxuria Baths: PHP / Laravel / MySQL — Sanitaryware catalog and business website. Frontend + Backend development.

Complete portfolio of 16+ featured platforms available at: https://pratik-mandani.github.io/

EDUCATION

Diploma in Electronics & Communication Engineering

Passout: 2018

Gujarat Technological University (GTU)

LANGUAGES & STRENGTHS

Languages: Gujarati, Hindi, English

Core Strengths: Product Problem Solving, Hardware Troubleshooting, Cross-functional Coordination, System Integration, Debugging, Practical Engineering, Production Support, Client Communication, Technical Issue Resolution, Learning & Prototyping.